

Problem 10.35

Simplify the following expression:

$$\left(\frac{d\delta}{dv}\right)\left(\frac{dd}{di}\right).$$

Solution

Recall that

$$v = \frac{1}{1+i} \quad \text{and} \quad \delta = \ln(1+i).$$

Since $1+i = v^{-1}$, the force of interest can be written as

$$\delta = \ln(v^{-1}) = -\ln v.$$

Therefore,

$$\frac{d\delta}{dv} = -\frac{1}{v}.$$

The effective discount rate is

$$d = \frac{i}{1+i}.$$

Differentiating with respect to i gives

$$\begin{aligned} \frac{dd}{di} &= \frac{(1+i) - i}{(1+i)^2} \\ &= \frac{1}{(1+i)^2} \\ &= v^2. \end{aligned}$$

Consequently,

$$\begin{aligned} \left(\frac{d\delta}{dv}\right)\left(\frac{dd}{di}\right) &= \left(-\frac{1}{v}\right)v^2 \\ &= -v. \end{aligned}$$

Thus, the simplified expression is

$$\boxed{-v}.$$